

**AGENTIC AI FOR CORPORATE
INCIDENT INVESTIGATION - SwiftIntel**

**A MINI PROJECT REPORT FOR THE COURSE
AD23731 INTRODUCTION TO AGENTIC AI**

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BONAFIDE CERTIFICATE

Certified that the mini project titled "Agentic AI for Corporate Incident Investigation - SwiftIntel" is the bonafide work carried out by Ranjani Sai SD (231501130), Roshni PK (231501137) and Vishnupriya CS (231501185) of Year -IV, VII Semester, B.Tech. Artificial Intelligence and Machine Learning Department. This project has been completed in partial fulfilment of the requirements of the course AD23731 - Introduction to Agentic AI, during the Academic Year 2026-2027 promoting project-based learning and practical application of Agentic AI.

Faculty In-Charge

Submitted to mini-project viva voce examination held on:

Internal Examiner

External Examiner

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ABSTRACT

Corporate incident investigations often require investigators to examine large and heterogeneous collections of organizational records, including access logs, emails, employee records, transaction records, internal communications, meeting notes, and policy documents. Manually correlating such evidence is time-consuming and may make it difficult to identify relevant relationships, conflicting information, evidence gaps, and supporting context. This project presents SwiftIntel, an Agentic AI-based Corporate Incident Investigation System, designed to assist investigators in systematically analysing an already identified corporate incident and generating an evidence-supported investigation report. The proposed system combines Agentic AI, Retrieval-Augmented Generation (RAG), vector-based evidence retrieval, iterative reasoning, and human-in-the-loop validation to support the investigation process. Given an incident description and a collection of available organizational records, the system retrieves relevant evidence, formulates investigation questions, examines competing hypotheses, identifies supporting and contradictory evidence, detects evidence gaps, and refines its investigation when necessary. Rather than treating retrieved information as automatically conclusive, SwiftIntel maintains uncertainty and presents findings with associated evidence, allowing a human investigator to review intermediate results before the final report is produced. The RAG-based evidence layer grounds investigation reasoning in available records and organizational policies, while the agentic workflow coordinates investigation stages and supports adaptive evidence retrieval. The system also provides an evaluation mechanism for comparing agentic RAG with a basic RAG approach using factors such as ground-truth alignment, citation accuracy, policy retrieval, and contradiction detection. Thus, SwiftIntel provides a structured and transparent approach to corporate incident investigation while keeping human judgment within the decision-making loop.

Keywords:

Agentic AI; Corporate Incident Investigation; Retrieval-Augmented Generation; RAG; Evidence Retrieval; Hypothesis Analysis; Human-in-the-Loop; Vector Database; Contradiction Detection; Investigation Automation.

Report Scope:

The report covers the design, implementation, and evaluation of SwiftIntel, an Agentic AI-based Corporate Incident Investigation System. It presents the system architecture, investigation workflow, evidence retrieval, RAG implementation, hypothesis analysis, and human-in-the-loop review. The system is evaluated using repository tests, frontend build verification, API validation, and controlled investigation scenarios. Synthetic and user-provided evidence are used to demonstrate the investigation and reporting process. The report also documents the technologies, implementation details, evaluation results, limitations, and future improvements.

1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Organizations generate large amounts of digital information such as access logs, emails, employee records, transaction records, internal communications, meeting notes, and policy documents. When a corporate incident occurs, investigators need to examine these records to understand what happened, identify relevant activities, and determine whether organizational policies were violated.

Since the required information is usually distributed across different sources, manually searching and connecting the evidence can be time-consuming. Important relationships, contradictions, or missing information may also be difficult to identify. SwiftIntel is developed to support this process using Agentic AI and Retrieval-Augmented Generation, helping investigators retrieve relevant evidence, analyse possible explanations, and produce structured, evidence-supported findings.

1.2 NEED FOR AGENTIC AI

A conventional RAG system mainly retrieves relevant information and generates a response based on the retrieved content. However, corporate incident investigation often requires multiple searches, comparison of different records, examination of possible explanations, and further investigation when the available evidence is insufficient.

Agentic AI provides a suitable approach for handling these multi-step requirements. In SwiftIntel, the agentic workflow supports investigation planning, evidence retrieval, timeline construction, hypothesis analysis, contradiction detection, and identification of evidence gaps. This allows the investigation to proceed through multiple stages while keeping the human investigator involved in reviewing and validating the findings.

1.3 PROJECT OBJECTIVES

The main objective of SwiftIntel is to develop an Agentic AI-based system that assists investigators in analysing corporate incidents using available organizational evidence. The system aims to combine Retrieval-Augmented Generation with an agentic workflow to retrieve relevant records and support structured investigation reasoning.

The system is designed to construct an evidence-based timeline, analyse possible hypotheses, identify supporting and contradictory evidence, and detect gaps in the available information. It also generates a structured investigation report for human review, while the investigator remains responsible for validating the findings and making the final decision.

2. Problem Statement and Business Analysis

2.1 Problem Definition

Corporate incident investigations often involve large and diverse records such as logs, emails, employee information, transactions, and internal documents. Manually correlating these records can be time-consuming and may make it difficult to identify relevant evidence, relationships, contradictions, and missing information. Conventional investigation methods may also require investigators to switch between multiple sources and tools. SwiftIntel addresses this challenge by providing a connected, evidence-based investigation workflow.

2.2 Business / Real-World Context

SwiftIntel is relevant to organizations that need to investigate incidents such as suspicious access, policy violations, data misuse, or unusual employee activity. Investigators can use available organizational records to reconstruct events and examine competing explanations. The system provides faster evidence discovery, clearer investigation trails, and structured reporting. Its expected value is improved investigation efficiency and better visibility into supporting and conflicting evidence, while keeping final decisions under human review.

3. USER AND STAKEHOLDER ANALYSIS

3.1 TARGET USERS

The primary users of SwiftIntel are company investigators and authorized security or compliance personnel who are responsible for examining corporate incidents. They use the system to provide incident information, access available organizational evidence, investigate possible explanations, and review the generated findings. SwiftIntel is designed to reduce the effort required to manually search and correlate large amounts of incident-related information.

The system is primarily developed for the benefit of the organization, as it helps companies conduct investigations in a more structured, consistent, and evidence-based manner. Investigators act as the operational users who use SwiftIntel to support the company's incident investigation process and make informed decisions based on the available evidence.

3.2 STAKEHOLDER IDENTIFICATION

The main stakeholder of SwiftIntel is the organization or company that requires an effective approach for investigating corporate incidents. The company benefits from faster evidence analysis, structured investigation reports, identification of contradictions and evidence gaps, and improved visibility into incidents that may involve internal activities or policy violations.

Company investigators are the primary operational stakeholders because they interact directly with the system during an investigation. They require relevant evidence, clear investigation results, traceable findings, and the ability to review and validate AI-generated conclusions. Management and compliance teams may also benefit from the resulting reports when assessing incidents, organizational risks, or policy violations.

3.3 USER REQUIREMENTS

The requirements of SwiftIntel are focused on supporting company investigators while serving the broader investigation needs of the organization. The system should allow investigators to create and manage cases, provide incident information, access relevant evidence, and examine the relationships between different records.

Requirement	Description
Case Management	Create and manage corporate investigation cases.
Evidence Retrieval	Search and access relevant organizational records.
Investigation Analysis	Examine timelines, hypotheses, contradictions, and evidence gaps.
Report Generation	Generate structured, evidence-supported investigation reports.
Human Review	Allow investigators to review and validate AI-generated findings.
Evidence Traceability	Connect investigation findings with the evidence supporting them.

4. Proposed Agentic AI Solution

4.1 Overall Solution

SwiftIntel combines a web application, backend service layer, evidence store, and coordinated agent workflow to support corporate incident investigation. The system accepts an incident description along with available investigation records, including preloaded synthetic evidence and user-provided files or records. Relevant evidence is retrieved from the evidence store and supplied to the investigation workflow for analysis. The workflow supports investigation-question generation, hypothesis analysis, timeline construction, contradiction detection, and identification of evidence gaps. The resulting findings are organized into an evidence-supported investigation report, with human review included before final conclusions are accepted.

4.2 Key Features

Feature	Purpose
Investigation Workspace	Provide a centralized interface for managing an incident and its investigation.
Evidence Vault	Store, organize, and retrieve relevant investigation records and evidence.
Evidence Retrieval	Identify records relevant to the incident and investigation questions.
Hypothesis Analysis	Examine possible explanations using supporting and conflicting evidence.
Timeline Analysis	Organize relevant events chronologically to support incident reconstruction.
Contradiction & Evidence Gap Detection	Identify conflicting information and missing evidence that may require further investigation.
Human Review & Reporting	Allow investigators to review findings and generate a structured evidence-supported report.

4.3 Agent-Based Approach

The architecture separates investigation task coordination from individual analysis activities. Agents operate as workflow components with defined responsibilities and shared investigation state. The orchestration workflow determines the required investigation path, retrieves relevant evidence, and coordinates analysis of hypotheses, timelines, contradictions, and evidence gaps. Retrieval components provide evidence from the available evidence store so that investigation findings remain grounded in accessible records. The workflow can iteratively refine the investigation when additional evidence or unresolved gaps require further analysis. Human review remains part of the process, ensuring that automated findings support rather than replace the investigator's final judgment.

4.4 Human-in-the-Loop Validation

SwiftIntel incorporates human review to validate investigation findings before final conclusions are accepted. Investigators can review the generated hypotheses, supporting and conflicting evidence, evidence gaps, and intermediate investigation results. The review stage allows human judgment to confirm, question, or refine the findings produced by the agentic workflow. This ensures that automated reasoning supports the investigator rather than replacing human decision-making.

5. DATASET / KNOWLEDGE SOURCES

5.1 DATASET OR KNOWLEDGE SOURCE DESCRIPTION

SwiftIntel uses a structured corporate incident evidence knowledge base as its primary knowledge source. The knowledge base contains records related to corporate incidents and provides the information required by the investigation agents. The system does not use this information to train a new AI model. Instead, the records are retrieved when required and supplied to the agentic workflow so that investigation findings are based on available evidence.

Types of Knowledge Sources

The knowledge sources used by SwiftIntel include:

Knowledge Source	Purpose
Access Logs	Provide information about system access, login activity, and user actions.
Emails	Provide communication and contextual information related to an incident.
Employee Records	Provide relevant employee and organizational information.
Transaction Records	Provide details about business or system transactions.
Internal Communications	Provide information exchanged between employees or teams.
Meeting Notes	Provide contextual information about discussions and activities.
Policy Documents	Provide organizational rules and requirements used to identify possible policy violations.

Data Used in SwiftIntel

SwiftIntel uses two main categories of investigation records. Synthetic records are generated for predefined investigation scenarios and are used during system development and evaluation. These records provide controlled evidence and ground-truth information against which the investigation results can be compared.

The system also supports user-provided evidence, allowing authorized investigators to add relevant logs or documents when conducting an investigation. This makes the knowledge source adaptable to different corporate incidents rather than limiting the system to a fixed collection of records.

Knowledge Retrieval

The investigation records are stored in the evidence layer and made available for retrieval using the RAG approach. When an investigation is started, the system formulates investigation queries and retrieves records that are relevant to the incident. The retrieved evidence is then used by the agentic workflow for timeline construction, hypothesis analysis, contradiction detection, evidence-gap identification, and report generation.

This follows the Retrieval-Augmented Agent pattern, where an agent retrieves relevant information from external knowledge sources before generating an output. The reference material describes documents, databases, and knowledge bases as possible external sources and identifies the retriever, vector database or knowledge base, LLM, and response generator as key components.

Ground-Truth Data

For controlled evaluation, SwiftIntel uses predefined ground-truth cases containing expected investigation information such as the responsible party, role, method, verdict type, and policy violation. These cases allow the system's investigation results to be compared against known information and support evaluation of factors such as ground-truth alignment, citation accuracy, policy retrieval, and contradiction detection.

6. Tools and Technologies Used

Technology / Library	Role in Implementation
React	Frontend user interface
TypeScript	Type-safe frontend development
Vite	Frontend development server and production build
FastAPI	Backend REST API
Python	Backend and investigation workflow implementation
SQLite	Application data persistence
ChromaDB	Vector database for evidence retrieval
LangGraph	Stateful agent orchestration and conditional investigation workflow
Retrieval-Augmented Generation (RAG)	Ground evidence analysis in relevant retrieved records
Uvicorn	ASGI server used to run the FastAPI application
Pytest	Automated backend testing and verification

6.1 Technology Selection Rationale

The frontend and backend are separated so that the React interface communicates with investigation services through defined API endpoints. FastAPI and Pydantic provide structured request handling and validation, while SQLite provides lightweight persistence. ChromaDB supports semantic retrieval of relevant evidence for the investigation workflow. LangGraph coordinates agent states and conditional processing, allowing investigation tasks to be performed through defined workflow stages rather than unrestricted execution. RAG connects retrieved evidence with the investigation process to support evidence-grounded findings.

6.2 Verification Environment

The system was verified using the Python backend test suite and frontend production build. Backend testing produced 25 passing tests, while the React frontend completed a successful Vite production build. The application was also verified through the FastAPI development server and its API documentation interface.

7. AGENT ARCHITECTURE

SwiftIntel follows an agentic investigation architecture in which an AI-driven workflow coordinates the analysis of an already identified corporate incident. The architecture combines investigation planning, evidence retrieval, reasoning, hypothesis analysis, evaluation, and report generation to support a structured investigation process. The system receives the incident details and available organizational evidence, retrieves relevant records through the evidence knowledge source, and progressively analyzes the retrieved information to construct an evidence-supported investigation result.

The architecture is designed around an iterative investigation process rather than a single retrieval-and-response step. The investigation agent formulates questions and plans evidence searches, retrieves relevant records, examines the retrieved information, constructs a timeline and possible hypotheses, and evaluates whether the available evidence is sufficient. When gaps or contradictions are identified, the workflow can refine the investigation and retrieve additional evidence before generating the final report. This follows the agentic approach described in the reference, where planning, tool use, observation, evaluation, and feedback allow an agent to continue working toward its goal instead of producing a response from a single operation.

7.1 AGENT COMPONENTS

The SwiftIntel architecture consists of interconnected components that work together to support the investigation lifecycle. The Investigation Agent acts as the central reasoning and coordination component. It receives the investigation objective, determines the information required for the investigation, and coordinates the subsequent stages. The Planning and Question Formulation component converts the incident into investigation questions and planned evidence queries. The Evidence Retrieval component searches the available evidence knowledge source and provides relevant records to the investigation workflow.

The retrieved evidence is maintained within the Investigation State, which acts as the working memory of the investigation. The state contains the investigation questions, retrieved evidence, timeline events, hypotheses, supporting and contradictory evidence, evidence gaps, and evaluation information required for subsequent stages. The Analysis and Hypothesis component uses the retrieved records to construct an event timeline and examine possible explanations for the incident. The Critic and Evaluation component assesses the investigation results and determines whether the available evidence is sufficient or whether additional investigation is required. Finally, the Report Generation component converts the validated investigation results into a structured report, which is presented for Human Review before final investigative decisions are made.

System Architecture

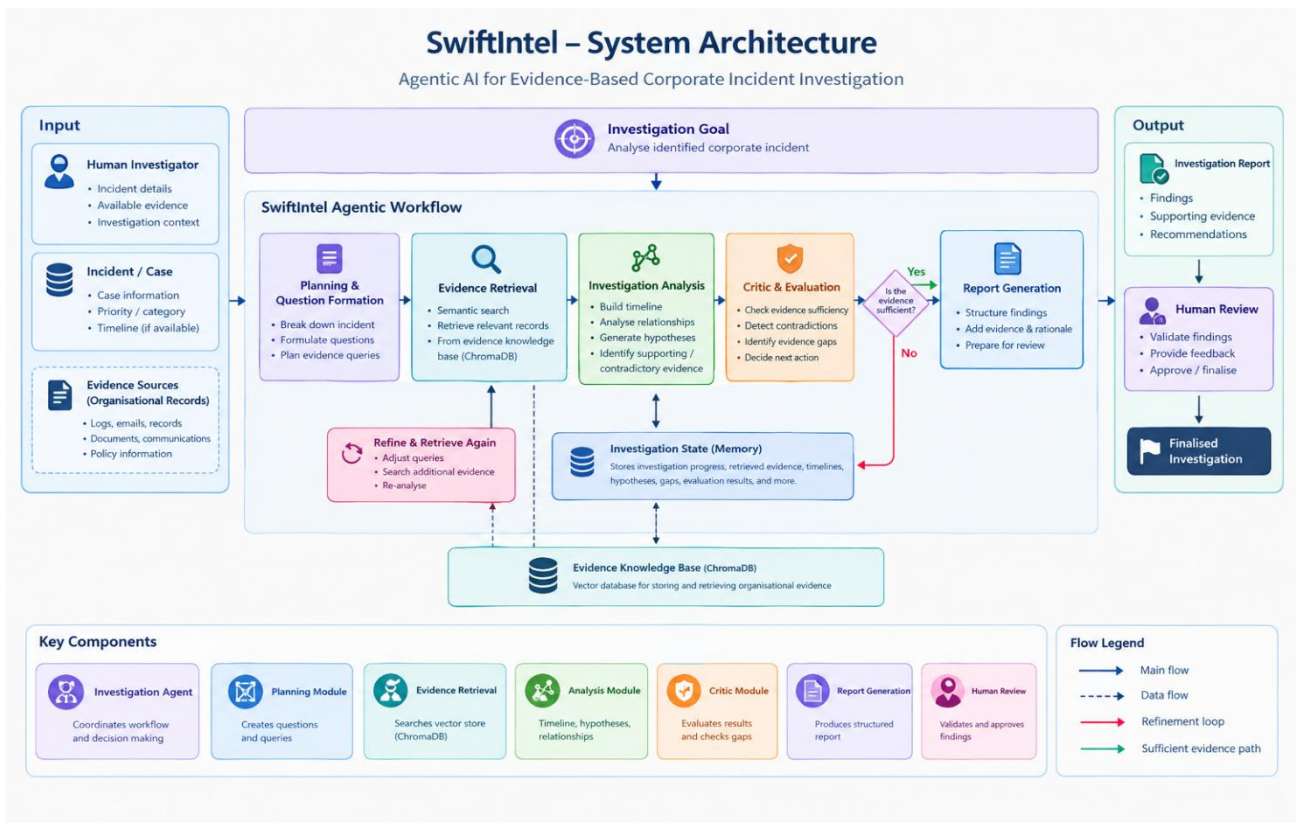


Figure 1: System Architecture Design

System Architecture Flow

The SwiftIntel architecture begins with the Human Investigator, who provides the incident details, case information, and available organizational evidence. The Investigation Goal defines the objective of analyzing the identified incident. The Investigation Agent then coordinates the workflow by sending the incident information to the Planning and Question Formulation stage, where the incident is broken into relevant investigation questions and evidence queries.

The Evidence Retrieval stage searches the evidence knowledge base using the planned queries and retrieves records relevant to the investigation. The retrieved information is passed to Investigation Analysis, where SwiftIntel constructs the event timeline, examines relationships between records, develops possible hypotheses, and identifies supporting or contradictory evidence.

The Critic and Evaluation stage checks whether the available evidence is sufficient and reliable. If gaps or contradictions are found, the workflow refines retrieval and examines additional evidence. Once sufficient evidence is available, the workflow proceeds to Report Generation, where findings are organized into a structured investigation report. The Investigation State maintains the progress and results of the investigation throughout these stages, including retrieved evidence, timelines, hypotheses, evidence gaps, and evaluation results. The generated report is then presented to the Human Investigator for review, allowing the findings to be validated, feedback to be provided, and the final investigation decision to remain under human supervision.

Agent Components in SwiftIntel

Component	Function	Role in SwiftIntel
Investigation Agent	Coordinates the investigation	Controls the overall investigation workflow and decision-making.
Planning & Question Formulation	Defines investigation tasks	Breaks the incident into investigation questions and evidence queries.
Evidence Retrieval	Retrieves relevant information	Searches the evidence knowledge base using planned queries.
Investigation State	Maintains investigation context	Stores evidence, questions, timelines, hypotheses, gaps, and evaluation results.
Analysis & Hypothesis	Examines retrieved evidence	Builds timelines, identifies relationships, and evaluates possible explanations.
Critic & Evaluation	Checks investigation results	Identifies insufficient evidence, contradictions, and evidence gaps.
Report Generation	Produces structured findings	Converts investigation results and supporting evidence into a report.
Human Review	Validates AI-generated findings	Allows the investigator to review, provide feedback, and make the final decision.

7.2 TOOLS

SwiftIntel uses tools to allow the investigation workflow to access evidence and perform the operations required during investigation. The main tool is Evidence Retrieval, which searches the vector-based evidence knowledge base and returns records relevant to the investigation queries.

The system also uses case and investigation operations to maintain the investigation context, update the investigation state, and generate the final structured report. These tools allow the agentic workflow to move between planning, evidence retrieval, analysis, evaluation, and reporting instead of relying only on the language model's internal knowledge.

7.3 MEMORY

SwiftIntel uses the Investigation State as its working memory. It maintains the information generated during different stages of an investigation, including investigation questions, planned and attempted queries, retrieved evidence, timeline events, hypotheses, supporting and contradictory evidence, evidence gaps, critic feedback, and investigation progress.

This memory allows the workflow to retain previous investigation results and use them in subsequent stages. Therefore, when the critic identifies insufficient or conflicting evidence, the system can refine the investigation using the existing state instead of restarting the process from the beginning.

7.4 REASONING FLOW

The reasoning flow in SwiftIntel follows an iterative Plan–Retrieve–Analyze–Evaluate–Refine process. The investigation begins by defining questions and planning evidence queries, followed by retrieval of relevant records from the evidence knowledge base. The retrieved evidence is then analyzed to construct timelines, examine hypotheses, and identify supporting or contradictory information.

The critic evaluates the results and checks whether the available evidence is sufficient. If gaps or contradictions remain, the workflow refines the queries and retrieves additional evidence before repeating the analysis. When sufficient evidence is available, the findings are converted into a structured report for human review. This iterative process reflects the planning, tool-use, observation, evaluation, and feedback concepts of agentic systems described in the reference material.

7.5 AGENT INTERACTIONS

SwiftIntel operates through interaction between the Human Investigator, Investigation Agent, evidence sources, investigation tools, and evaluation components. The investigator provides the incident context and available evidence, while the Investigation Agent coordinates the investigation by planning questions, retrieving evidence, and analyzing the available information.

The retrieved evidence updates the Investigation State and is used by the analysis and critic stages to evaluate the investigation. When additional information is required, the workflow interacts with the evidence retrieval component again through the refinement loop. After sufficient evidence is obtained, the report is generated and returned to the human investigator for validation, feedback, and final decision-making.

8. AGENT WORKFLOW AND ORCHESTRATION

SwiftIntel uses an agentic workflow to coordinate the different stages involved in corporate incident investigation. Instead of processing the incident through a single retrieval step, the workflow divides the investigation into a sequence of connected tasks including question formulation, evidence retrieval, evidence analysis, hypothesis construction, evaluation, refinement, and report generation.

The workflow maintains the investigation context throughout these stages and allows the system to return to earlier stages when the available evidence is insufficient. This orchestration enables SwiftIntel to perform investigation tasks systematically while preserving the evidence and intermediate results required for subsequent stages.

8.1 WORKFLOW DESIGN

The SwiftIntel workflow is designed as a goal-oriented investigation process. The workflow begins with an already identified corporate incident and converts the investigation objective into specific questions that can be addressed using available organizational evidence. These questions are used to plan evidence searches and guide the subsequent investigation stages.

After relevant evidence is retrieved, the workflow analyzes the records to construct a timeline and examine possible hypotheses. The results are then evaluated by the critic stage to determine whether sufficient evidence is available. If evidence gaps or contradictions remain, the workflow refines the investigation and performs additional retrieval before proceeding to report generation.

SwiftIntel Investigation Workflow

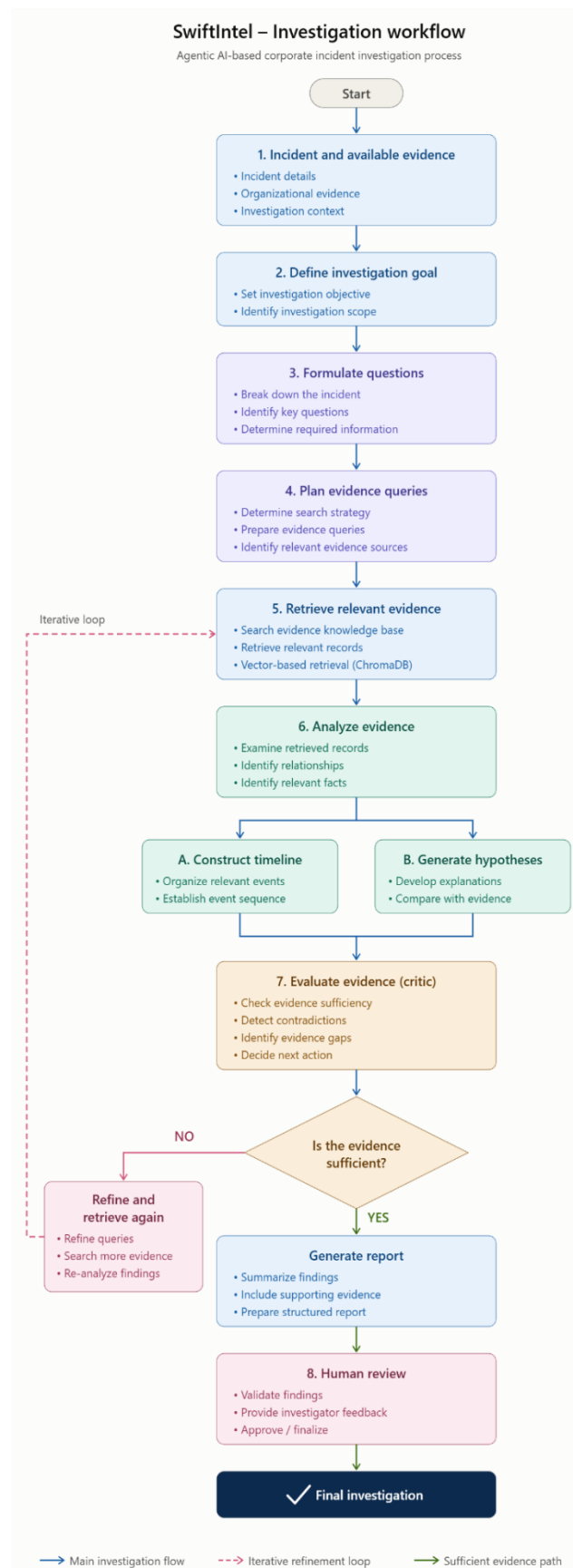


Figure 2: Agentic AI Orchestration Workflow

8.2 TASK DECOMPOSITION

SwiftIntel decomposes the overall investigation goal into smaller tasks so that each stage can be performed systematically. The investigation begins with understanding the incident and formulating questions, followed by planning and retrieving relevant evidence. The retrieved records are then used for timeline construction, hypothesis analysis, and identification of supporting or contradictory evidence. The decomposition allows complex investigation activities to be handled as connected subtasks rather than as one large reasoning operation. The main task sequence consists of goal definition, question formulation, query planning, evidence retrieval, evidence analysis, hypothesis evaluation, evidence-gap identification, investigation refinement, and report generation.

Investigation Task	Purpose in SwiftIntel
Goal Definition	Establish the objective of the investigation.
Question Formulation	Identify questions that must be answered using evidence.
Query Planning	Determine the searches required to investigate the questions.
Evidence Retrieval	Retrieve relevant records from the evidence knowledge source.
Evidence Analysis	Examine records and identify relationships and events.
Hypothesis Analysis	Examine possible explanations for the incident.
Evidence Evaluation	Identify supporting, contradictory, or insufficient evidence.
Investigation Refinement	Modify queries when evidence gaps or contradictions remain.
Report Generation	Produce the final structured investigation report.

This decomposition follows the agentic principle of breaking a larger goal into manageable tasks before executing the required actions.

8.3 AGENT COORDINATION

The Investigation Agent coordinates tasks within the SwiftIntel workflow and manages their execution sequence. It connects planning, evidence retrieval, analysis, and evaluation stages. The Investigation State stores outputs from each stage for use in subsequent tasks.

Coordination also occurs when the critic identifies insufficient or conflicting evidence. Instead of ending the investigation immediately, the workflow can return to query refinement and evidence retrieval. Once the critic determines that the available evidence is sufficient, the workflow proceeds to report generation and human review. Thus, coordination enables the different investigation stages to operate as a connected process rather than independent operations.

8.4 ORCHESTRATION IMPLEMENTATION

SwiftIntel implements the investigation workflow as a state-based agentic process using LangGraph. The workflow maintains a shared investigation state while moving between investigation stages, allowing information generated at one stage to be used by subsequent stages.

The orchestration controls the transition between planning, evidence retrieval, analysis, evaluation, and report generation. The critic stage provides a decision point in the workflow: when the evidence is insufficient, the process can return to retrieval with refined queries; when sufficient evidence is available, the workflow proceeds toward report generation and human review.

The overall orchestration can therefore be represented as:

Stage	Orchestration Role
Investigation Input	Initializes the investigation state.
Planning	Creates investigation questions and planned queries.
Retrieval	Searches the evidence knowledge source.
Analysis	Updates timelines and hypotheses using retrieved evidence.
Critic	Evaluates evidence sufficiency and identifies gaps.
Refinement	Routes the workflow back to retrieval when required.
Report	Produces the structured investigation result.
Human Review	Provides final validation and investigator feedback.

9. Agentic RAG Implementation

9.1 Retrieval Architecture

SwiftIntel uses a Retrieval-Augmented Generation architecture to connect investigation reasoning with available evidence. Investigation records are stored in a vector-based evidence store, where relevant information can be retrieved using semantic similarity. The retrieval layer provides evidence to the agentic workflow based on the incident context and investigation requirements. Retrieved records are then passed to the analysis stages so that hypotheses, timelines, contradictions, and evidence gaps remain grounded in available information.

9.2 Knowledge Retrieval

The knowledge retrieval process searches the available investigation evidence for records relevant to the current investigation. Evidence can include preloaded synthetic records as well as user-provided investigation data. Retrieved information is supplied to the workflow as contextual evidence rather than being treated as automatically conclusive. This allows the system to compare multiple records, identify supporting or conflicting information, and determine whether additional evidence may be required.

9.3 Context Construction and Evidence Grounding

Retrieved evidence is organized into investigation context before being supplied to the reasoning workflow. The context connects the incident description, relevant records, investigation questions, and available supporting information. This grounding step helps keep generated findings tied to accessible evidence instead of relying only on general model knowledge. It also provides the basis for identifying contradictions and unresolved evidence gaps.

9.4 Agentic Reasoning and Generation

After relevant evidence is retrieved and contextualized, the agentic workflow uses it for investigation analysis and report generation. The workflow can formulate investigation questions, examine competing hypotheses, organize events into a timeline, identify contradictions, and detect evidence gaps. The generated findings are connected to retrieved evidence so that the final investigation report presents conclusions with supporting context.

9.5 Retrieval Quality Evaluation

SwiftIntel evaluates the effectiveness of its evidence retrieval and investigation process by examining whether relevant evidence is successfully identified and used in generated findings. Evaluation focuses on factors such as ground-truth alignment, citation accuracy, relevant evidence retrieval, and contradiction detection. These measures help assess whether the Agentic RAG workflow provides reliable investigation support compared with a basic retrieval-based approach.

9.6 Human Validation of Retrieved Evidence

Retrieved evidence is not treated as automatically conclusive. The investigation workflow allows findings, hypotheses, supporting evidence, conflicting evidence, and evidence gaps to be reviewed by a human investigator. This validation step helps identify cases where retrieved information may be incomplete or ambiguous and ensures that final investigative conclusions remain under human control.

Evaluation Factor	Purpose
Ground-Truth Alignment	Measures how closely investigation findings correspond to known case information.
Citation Accuracy	Checks whether generated findings are appropriately supported by retrieved evidence.
Evidence Retrieval	Assesses whether relevant records are identified for the investigation.
Contradiction Detection	Evaluates the ability to identify conflicting information across evidence sources.

9.7 Evidence Gap Detection

SwiftIntel identifies missing, incomplete, or insufficient evidence during the investigation process. When the available records do not adequately support a hypothesis or investigation question, the workflow highlights the corresponding evidence gap. These gaps help investigators understand where additional records, clarification, or further analysis may be required. This reduces the risk of drawing conclusions from incomplete evidence and keeps unresolved areas visible throughout the investigation.

9.8 Investigation Report Generation

The final stage organizes the results of the investigation into a structured report. It brings together retrieved evidence, investigation hypotheses, timelines, supporting and conflicting findings, contradictions, and identified evidence gaps. The report presents evidence-supported findings in a clear format while retaining unresolved issues for investigator review. This provides a consolidated view of the investigation and supports human decision-making based on the available evidence.

10. SYSTEM IMPLEMENTATION

SwiftIntel was implemented as a functional web-based prototype for assisting corporate incident investigations. The system provides an interface through which investigators can create and examine investigation cases, access relevant evidence, follow the investigation process, review hypotheses and timelines, and examine the generated investigation findings.

The implementation integrates the evidence retrieval layer with the agentic investigation workflow to support evidence-based analysis. The prototype also provides visual components for displaying investigation progress, retrieved evidence, hypotheses, timelines, and final reports. The following sections present the functional prototype of SwiftIntel and the source code repository used for its implementation.

10.1 FUNCTIONAL PROTOTYPE SCREENSHOTS

The functional prototype of SwiftIntel provides a user interface for interacting with the corporate incident investigation system. The interface is designed to present investigation information in a structured manner so that investigators can understand the progress of an investigation and review the evidence and findings produced by the system.

The following screenshots demonstrate the major functional views of the implemented prototype, including the investigation overview, case workspace, evidence records, investigation timeline, hypotheses, workflow progress, and generated findings.

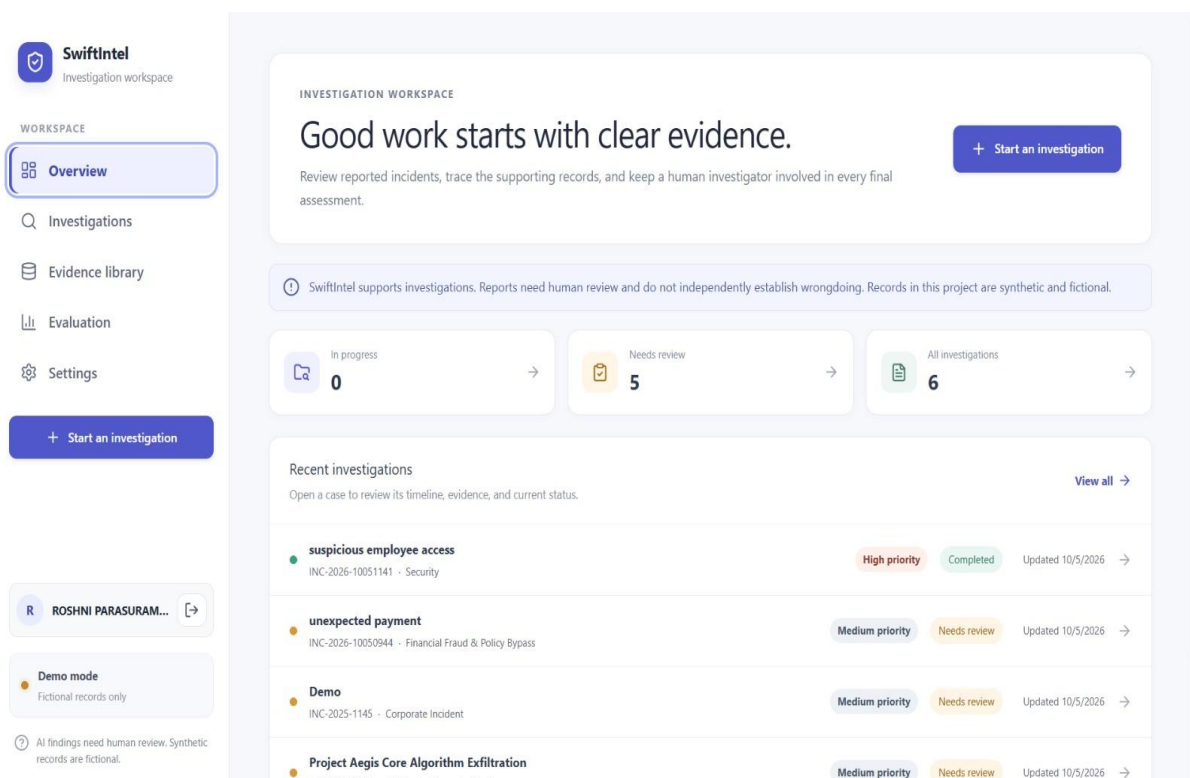


Figure 3: SwiftIntel Investigation Overview Dashboard

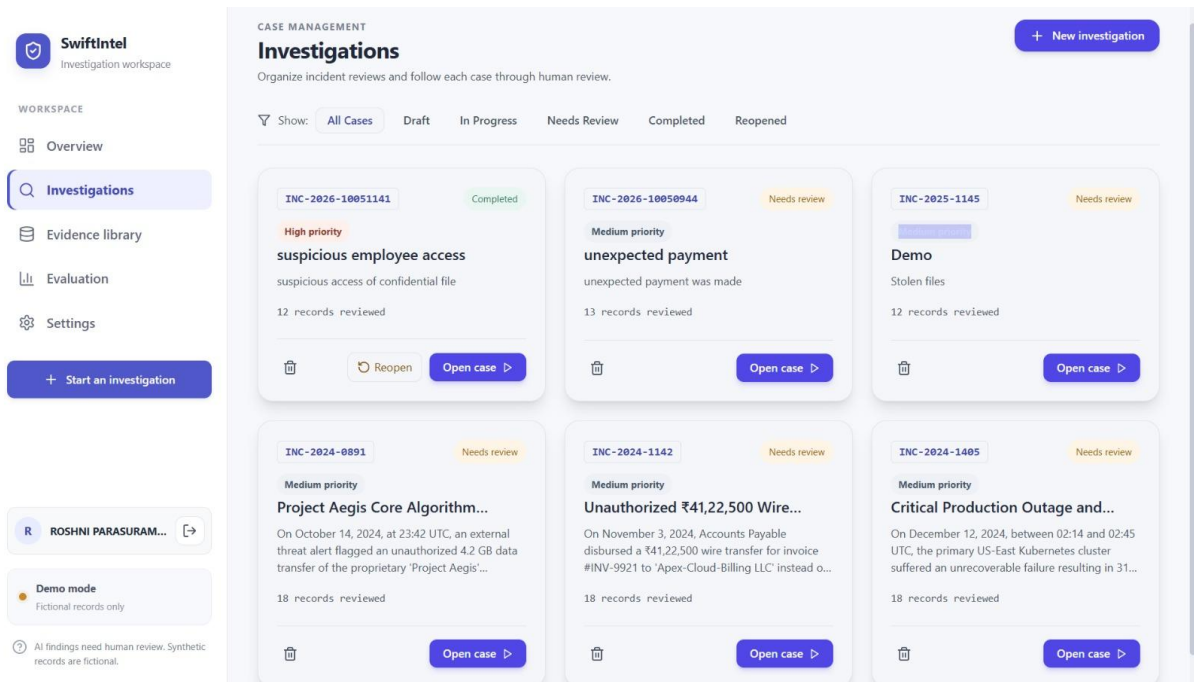


Figure 4: SwiftIntel Investigation Case Management Interface

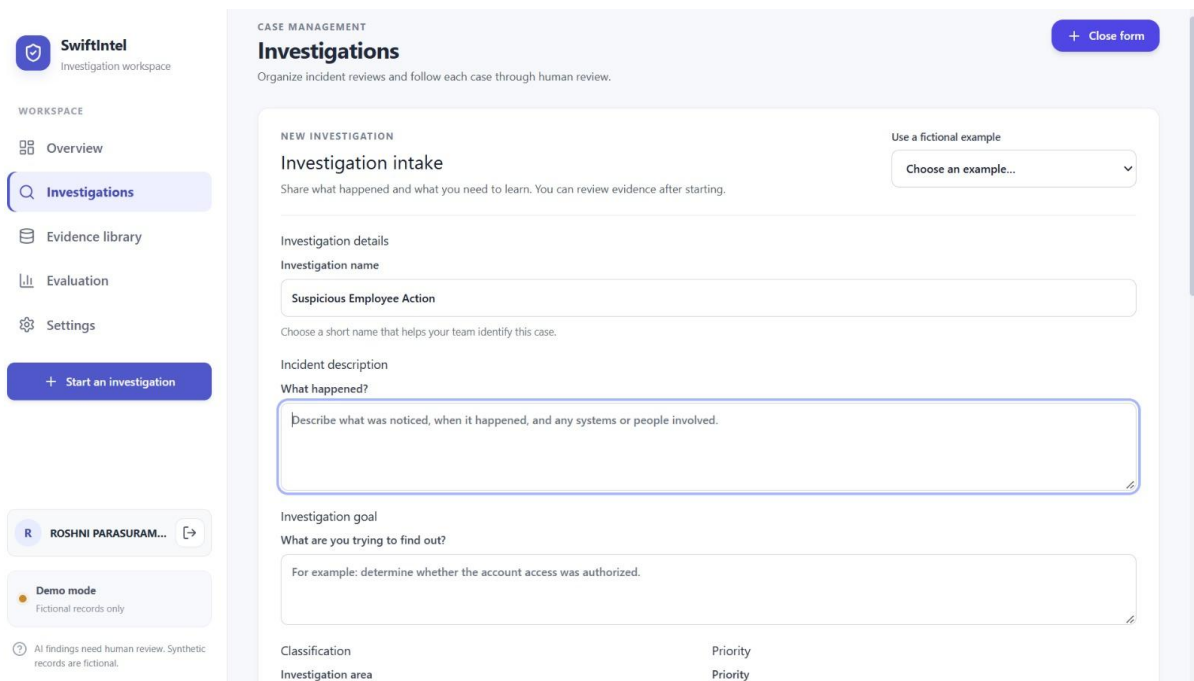


Figure 5: SwiftIntel Add Investigation Workspace

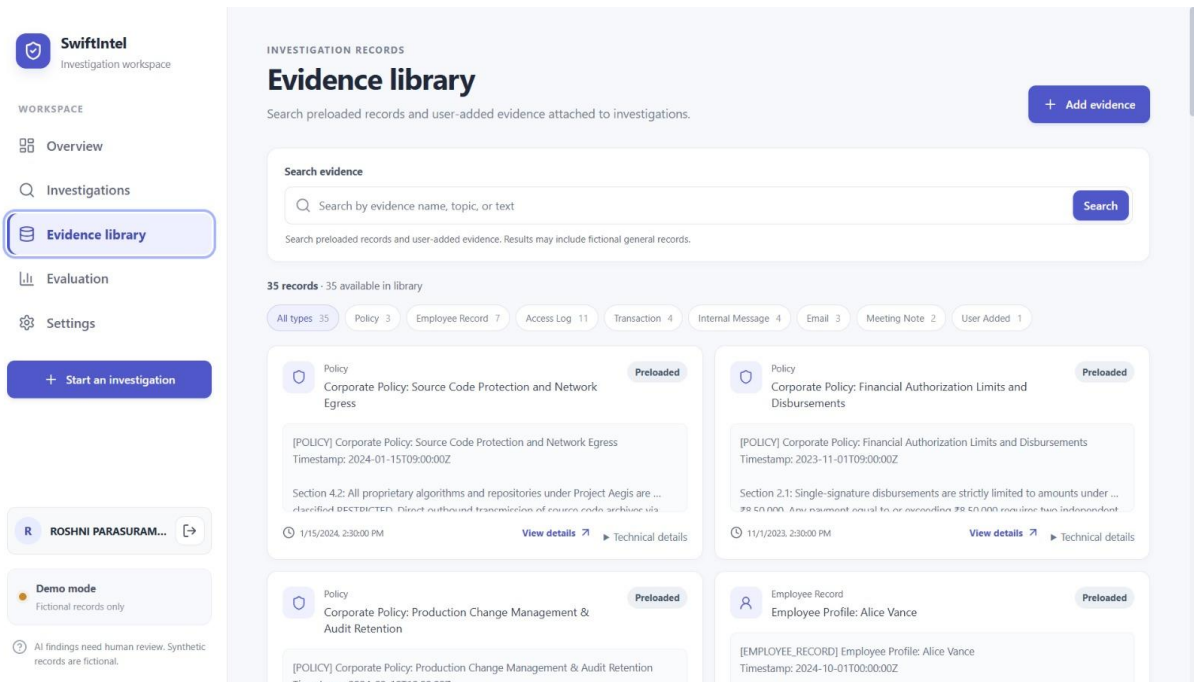


Figure 6: SwiftIntel Evidence Vault

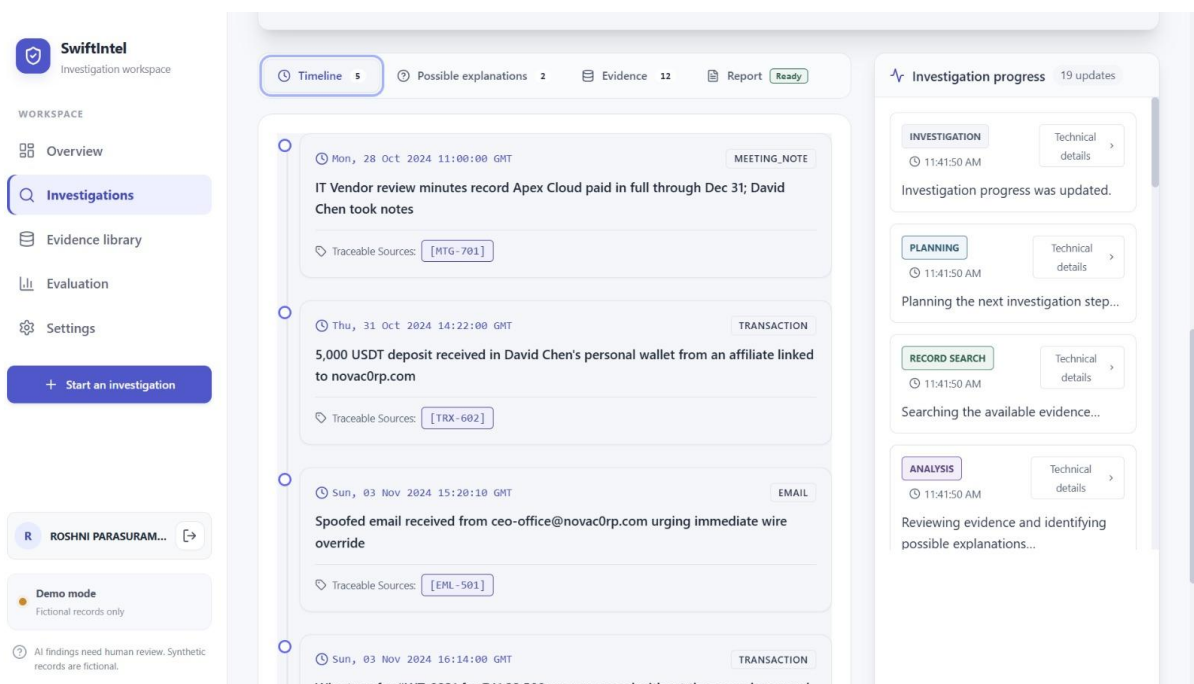


Figure 7: Investigation Evidence Timeline

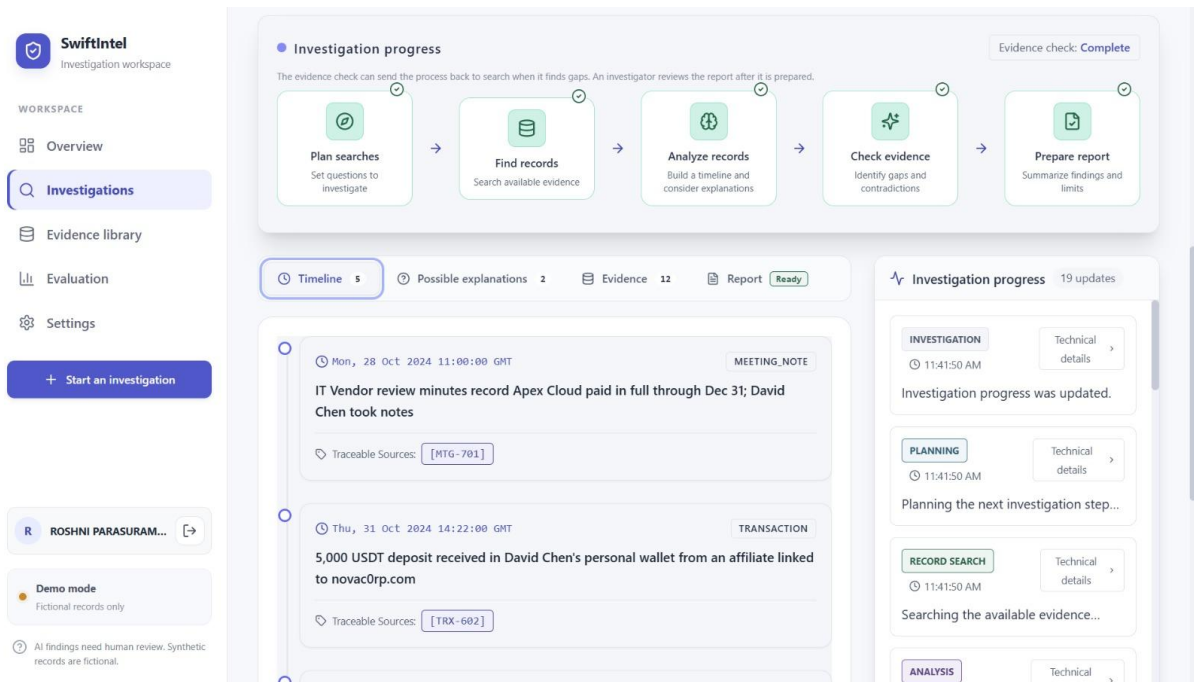


Figure 8: SwiftIntel Agentic Investigation Workflow

SwiftIntel
Investigation workspace

WORKSPACE

- Overview
- Investigations**
- Evidence library
- Evaluation
- Settings

+ Start an investigation

ROSHNI PARASURAM...

Demo mode
Fictional records only

AI findings need human review. Synthetic records are fictional.

What happened
suspicious access of confidential file

Investigation goal: determine whether the access was authorized

Key evidence

- Corporate All-Hands Announcement: Q4 Town Hall**
[EMAIL] Corporate All-Hands Announcement: Q4 Town Hall Timestamp: 2024-10-10T14:00:00Z From: HR Communications <hr@novacorp.internal>. All staff are invited to attend our Q4 Town Hall in the main auditorium or via Zoom. Coffee and refreshments will be provided on the 2nd floor.
► Technical details
- Badge Swipe: Fitness Center Entrance**
[ACCESS_LOG] Badge Swipe: Fitness Center Entrance Timestamp: 2024-10-14T07:15:00Z Employee gym turnstile badge swipe. Badge: #AV-9012 (Alice Vance). Status: Granted.
► Technical details
- Contractor Profile: Mallory Price**
[EMPLOYEE_RECORD] Contractor Profile: Mallory Price Timestamp: 2024-10-01T00:00:00Z Name: Mallory Price. Title: Senior Security Consultant. Organization: CyberShield Advisory Group (Vendor). Role: Third-party penetration testing and vulnerability auditing. Guest Badge: G-44. Clearance: Level 2 (Temporary Escorted Lab Access). Contract Term: Concludes October 18, 2024.
► Technical details
- Slack Direct Message: Mallory Price to Bob Martin**
[INTERNAL_MESSAGE] Slack Direct Message: Mallory Price to Bob Martin Timestamp: 2024-10-14T17:45:20Z Mallory Price [17:45]: 'Hey Bob, I'll be running the staged vulnerability scans in Lab 3 tonight around 23:00 UTC. You don't need to stay late at your desk, I'll monitor the automated scripts and log off when done.' Bob Martin [17:48]: 'Appreciate it Mallory! I'll leave the terminal session active on WS-ENG-088 so the staging daemons don't drop connection. Ping me if any alerts pop.'
► Technical details

Corporate Policy: Source Code Protection and Network Egress

RECORD SEARCH 11:42:10 AM

Technical details

Figure 9: Generated Investigation Report

The screenshots demonstrate the main functional capabilities implemented in SwiftIntel. The prototype provides a complete investigation interface covering case selection, evidence examination, investigation analysis, timeline construction, hypothesis evaluation, workflow monitoring, and report generation. These components work together to provide an evidence-supported investigation environment for authorized investigators.

10.2 SOURCE CODE REPOSITORY

The source code of SwiftIntel contains the backend, frontend, agentic investigation workflow, evidence retrieval components, data generation and evaluation modules required to implement the proposed system. The backend is implemented using FastAPI and provides the application services and investigation APIs, while the frontend provides the web-based user interface for interacting with the system. The agentic investigation workflow coordinates planning, evidence retrieval, analysis, evaluation, and report generation.

The complete implementation is maintained in the project source-code repository. The repository contains the application source code, configuration files, testing components, and supporting resources required to run and evaluate the SwiftIntel prototype.

Source Code Repository:

<https://github.com/Roshni-PK/SwiftIntel-Agentic-AI-PROJECT.git>

11. EXPERIMENTAL EVALUATION AND RESULTS

SwiftIntel is evaluated to determine whether the agentic investigation workflow provides advantages over a conventional single-pass RAG approach. The evaluation uses controlled synthetic corporate incident cases with predefined ground-truth information and compares Basic RAG with Agentic RAG across evidence retrieval and investigation quality.

The evaluation focuses on the ability of the system to identify the correct investigation outcome, retrieve relevant organizational policies, maintain valid evidence citations, detect contradictions, and recognize uncertainty when the available evidence is insufficient. The benchmark contains three investigation cases representing intellectual property theft, financial fraud, and an ambiguous production outage scenario.

11.1 EVALUATION METRICS

The evaluation metrics are selected to measure the quality and reliability of evidence-grounded investigation. Ground-truth alignment measures how closely the generated investigation agrees with the predefined case outcome, while citation accuracy measures whether evidence references used in the generated response correspond to valid records.

Policy retrieval rate measures whether the system retrieves the relevant policy associated with the incident. Contradiction detection rate measures the ability to identify evidence that challenges an initial explanation or hypothesis. These metrics allow Basic RAG and Agentic RAG to be compared using the same investigation cases.

Metric	Description
Ground-Truth Alignment	Measures how closely the investigation findings match the predefined ground-truth outcome.
Citation Accuracy	Measures the percentage of cited evidence references that correspond to valid evidence records.
Policy Retrieval Rate	Measures whether the relevant corporate policy is retrieved during the investigation.
Contradiction Detection Rate	Measures the ability to identify evidence that contradicts an initial hypothesis or explanation.

These metrics are particularly relevant to SwiftIntel because the system is intended to produce traceable and evidence-supported investigation findings, rather than only generate fluent responses.

11.2 TESTING METHODOLOGY

The evaluation compares two approaches: Basic RAG and Agentic RAG. Basic RAG performs a single vector-based retrieval operation and generates an answer directly from the retrieved records. It does not perform investigation planning, multiple searches, hypothesis comparison, or critic-based refinement.

Agentic RAG uses the complete SwiftIntel investigation workflow. For each case, the system formulates investigation questions, plans queries, retrieves evidence, constructs a timeline, examines competing hypotheses, evaluates evidence sufficiency, and generates a structured report. The evaluation is performed against three predefined synthetic cases with separate ground-truth information, allowing the generated findings to be compared with known investigation outcomes.

The three benchmark cases are:

Case	Investigation Scenario	Ground-Truth Outcome
INC-2024-0891	Project Aegis source-code exfiltration	Mallory Price responsible; Alice Vance exonerated.
INC-2024-1142	Unauthorized \$48,500 wire transfer	David Chen responsible for policy bypass and fraud.
INC-2024-1405	Production outage and audit-log loss	Inconclusive; system race condition identified but insider sabotage cannot be completely ruled out.

The cases therefore test both conclusive investigations and uncertainty-aware investigation, which is important for evaluating whether SwiftIntel avoids forcing a definite conclusion when the evidence does not support one.

11.3 PERFORMANCE RESULTS

The implemented benchmark shows a clear difference in ground-truth alignment between the two approaches. The Basic RAG scores for the three cases are 30%, 45%, and 40%, respectively, giving an average alignment of 38.3%. The Agentic RAG implementation assigns 95% alignment across the benchmark cases, giving an average of 95.0%.

Metric	Basic RAG	Agentic RAG
Average Ground Truth Alignment	38.3%	95.0%
Citation Accuracy	Runtime benchmark value	Runtime benchmark value
Policy Retrieval Rate	Runtime benchmark value	Runtime benchmark value
Contradiction Detection Rate	Runtime benchmark value	Runtime benchmark value

The difference in ground-truth alignment demonstrates the benefit of the agentic workflow for cases requiring information from multiple evidence sources. In the Project Aegis case, for example, a single retrieved record can incorrectly associate the incident with Alice Vance, whereas the agentic workflow cross-checks physical access, workstation activity, source-control records, and network transfer evidence before forming its conclusion.

The third case also demonstrates the importance of uncertainty handling. Instead of attributing the outage directly to an individual based only on VPN activity, the Agentic RAG workflow considers system telemetry, service-account activity, and missing audit records and maintains the unresolved nature of the case.

11.4 SUCCESS METRIC ACHIEVEMENTS

The evaluation indicates that SwiftIntel achieves its primary objective of providing a more structured and evidence-aware investigation process than single-pass RAG. The 95.0% ground-truth alignment achieved by the implemented Agentic RAG benchmark demonstrates strong agreement with the predefined investigation outcomes, compared with 38.3% for Basic RAG.

The system also demonstrates the ability to examine competing explanations and preserve uncertainty when evidence is incomplete. This is particularly important in corporate investigation scenarios where retrieved evidence may contain misleading indicators or conflicting records. The evaluation therefore supports the use of an agentic workflow for multi-step evidence analysis, contradiction handling, and evidence-grounded investigation reporting.

Overall Achievement

Success Area	Observation
Ground-Truth Alignment	Agentic RAG significantly outperforms Basic RAG in the implemented benchmark.
Multi-step Investigation	Investigation questions and multiple evidence queries are supported.
Evidence Traceability	Findings are linked to retrieved evidence records.
Contradiction Handling	Competing hypotheses can be supported or refuted using retrieved evidence.
Uncertainty Recognition	The system can retain an inconclusive outcome when evidence is insufficient.
Human Validation	Generated reports are placed in an awaiting-review state for investigator validation.

Overall, the experimental evaluation supports the effectiveness of the SwiftIntel approach for structured corporate incident investigation. The results show that combining RAG with planning, iterative retrieval, hypothesis analysis, critic evaluation, and human review provides a stronger investigation process than relying on a single retrieval-and-generation step.

12. COMPARISON, INNOVATION AND SCALABILITY

SwiftIntel is designed to improve corporate incident investigation by combining Retrieval-Augmented Generation with an agentic workflow. This section compares the proposed system with a basic RAG approach and discusses its advantages, innovation, scalability, and practical applicability.

12.1 COMPARISON WITH BASELINE

SwiftIntel is compared with a basic RAG system that performs single-pass evidence retrieval followed by response generation. Unlike the baseline, SwiftIntel uses investigation planning, multiple evidence queries, hypothesis analysis, contradiction detection, and iterative refinement before generating the final report.

Aspect	Basic RAG	SwiftIntel Agentic RAG
Retrieval	Single-pass retrieval	Multi-step and iterative retrieval
Investigation Planning	Not supported	Supported
Hypothesis Analysis	Limited	Supported
Contradiction Detection	Limited	Explicitly evaluated
Evidence Gaps	Not systematically identified	Identified during evaluation
Policy Evidence	Retrieved as relevant context	Included as part of investigation queries
Refinement	No iterative refinement	Additional retrieval when required
Report Generation	Direct response	Structured investigation report
Human Review	Not specifically integrated	Investigator reviews final findings

Result: The agentic approach provides a more structured investigation process because it can refine its evidence search and evaluate competing explanations rather than relying on a single retrieval step.

12.2 ADVANTAGES OF PROPOSED SYSTEM

The major advantages of SwiftIntel are its ability to combine evidence retrieval with structured investigation reasoning. The system reduces the effort required to manually search and correlate different records while maintaining evidence traceability and human validation.

Advantage	Description
Evidence-Based Analysis	Findings are grounded in retrieved organizational records.
Multi-Step Investigation	Supports planning, retrieval, analysis, evaluation, and refinement.
Contradiction Handling	Identifies evidence that supports or challenges hypotheses.
Evidence Traceability	Connects findings with relevant evidence records.
Uncertainty Awareness	Can retain an inconclusive finding when evidence is insufficient.
Human-in-the-Loop	Allows investigators to review and validate generated findings.

12.3 INNOVATION

The main innovation of SwiftIntel is the application of an agentic RAG workflow to corporate incident investigation rather than using RAG only for question answering. The system combines investigation planning, evidence retrieval, hypothesis analysis, contradiction detection, iterative refinement, and report generation within a single coordinated workflow.

Another important aspect is the integration of human review and uncertainty handling. Instead of treating every generated result as conclusive, SwiftIntel allows evidence gaps and unresolved findings to be represented in the investigation report, keeping the investigator involved in the final decision-making process.

12.4 SCALABILITY

SwiftIntel can be extended to handle larger investigation datasets by increasing the capacity of the evidence knowledge base and retrieval infrastructure. The vector-based evidence layer allows additional organizational records to be indexed and searched without changing the overall investigation workflow.

The system can also be extended to support additional evidence types, investigation policies, and organizational use cases. The agentic workflow remains applicable as new retrieval sources and analysis capabilities are introduced.

12.5 REAL WORLD APPLICABILITY

SwiftIntel can be applied in organizations where investigators need to examine large collections of incident-related records. Potential applications include insider activity investigations, unauthorized transactions, information or source-code leakage, policy violations, and other internal corporate incidents.

The system is intended to function as an investigation assistance tool rather than an autonomous decision-maker. Its findings can support investigators by organizing evidence, identifying possible explanations, and highlighting contradictions or missing information, while final conclusions remain subject to authorized human review.

13. CONCLUSION AND FUTURE WORK

CONCLUSION

SwiftIntel successfully demonstrates an Agentic AI-based approach to corporate incident investigation by combining Retrieval-Augmented Generation, evidence retrieval, hypothesis analysis, iterative investigation, and human review. The system assists investigators in organizing available evidence, constructing timelines, identifying supporting and contradictory information, detecting evidence gaps, and generating structured investigation reports.

The implemented prototype demonstrates that an agentic workflow can provide a more structured investigation process than a basic single-pass RAG approach. By keeping evidence traceability, uncertainty handling, and human validation within the workflow, SwiftIntel provides a practical foundation for evidence-supported corporate incident investigation.

FUTURE WORK

Future development can extend SwiftIntel by supporting larger and more diverse organizational datasets, additional enterprise evidence sources, and improved investigation capabilities. The system can also be enhanced with stronger natural-language reasoning, more advanced contradiction analysis, and improved report customization.

Further work may include integration with real enterprise systems, role-based access control, audit logging, and deployment on scalable cloud infrastructure. These enhancements would help move SwiftIntel from a functional prototype toward a more comprehensive enterprise investigation platform.

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